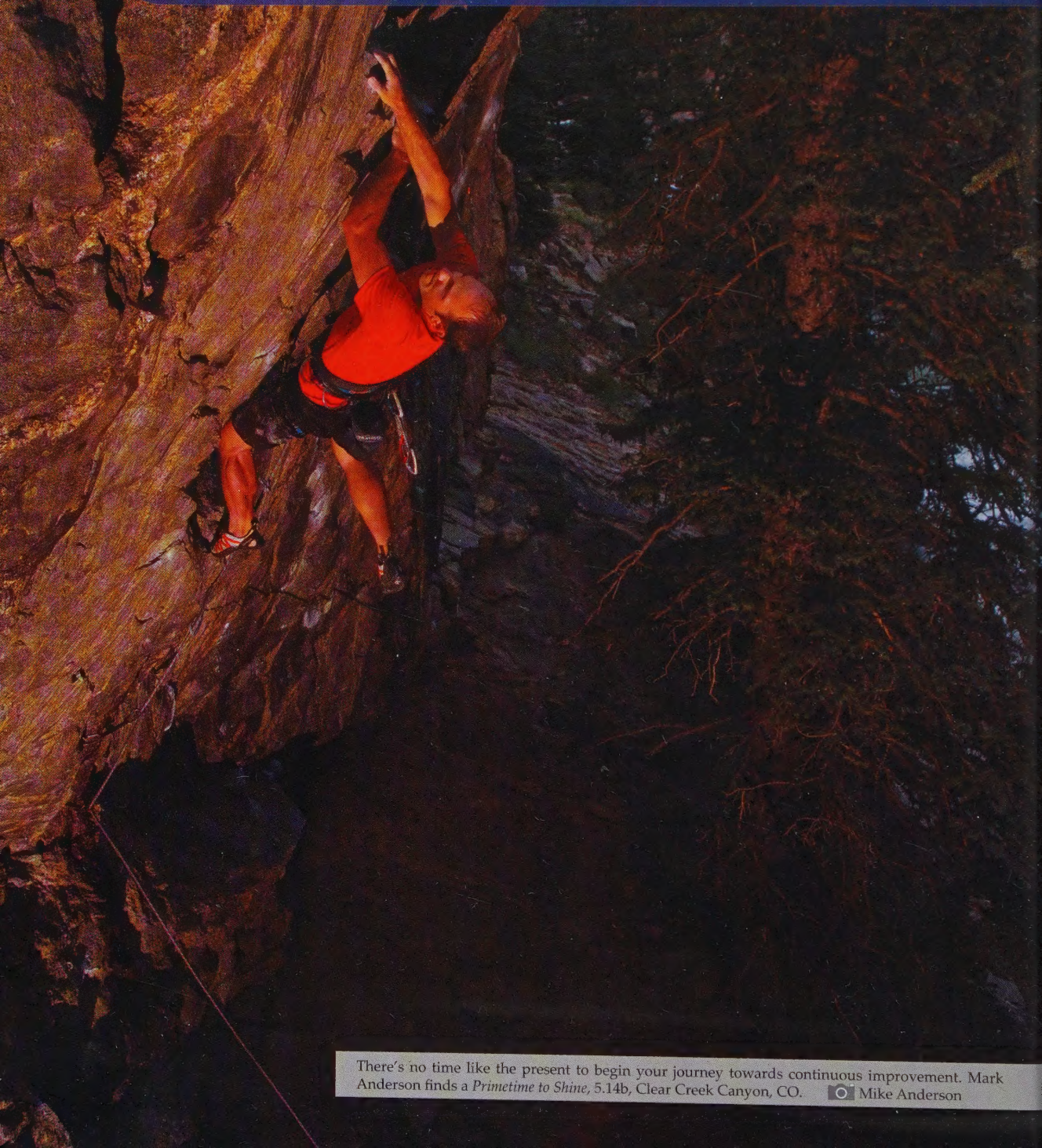


INTRODUCTION



There's no time like the present to begin your journey towards continuous improvement. Mark Anderson finds a *Primetime to Shine*, 5.14b, Clear Creek Canyon, CO. 📷 Mike Anderson

CHAPTER

1

"Whatever you can do or dream you can, begin it. Boldness has genius, power and magic in it."

— Goethe

There are some routes that just call to us like Homer's Sirens. A laser-cut arête, a towering line of crisp edges, an endless splitter or an epic citadel of Sierra granite. The truly beautiful lines captivate our consciousness and demand the very most from our physical abilities. These lines keep us awake at night, make our palms sweat, and occupy our attention to the exclusion of all else. And they demand to be climbed. Satisfaction can only be achieved through success. These lines require us to raise our game to new heights, literally and figuratively. But how to do so? Inspiration lies around every buttress. Hard work and commitment can be summoned for the right objective, but climbing is complex, as is the human machine, and finding the right path to reach your goals is not simple.

THE CASE FOR TRAINING

TRAINING DEFINED

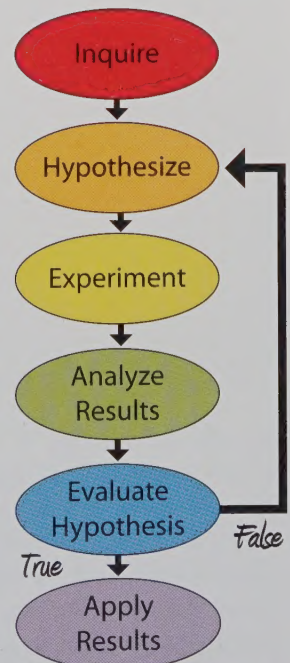
So what is training anyway? In terms of climbing, and for the purposes of this book, *training* is defined as an approach to *improve climbing performance* that is *systematic*, *disciplined*, and *science based*. Thus, this training program is focused on the singular goal of producing improved climbing performance. Other attributes of a training program, such as how much sweat it produces or how exhausting it feels, are entirely irrelevant; only the effect on climbing performance counts. When faced with a novel training method, every climber should ask: Will this improve my climbing performance?

A training program is *systematic* when it is premeditated and organized in a logical way, leading to a defined goal. Thus, it should be designed deliberately to transform the body from a known starting point to-

ward a specific, improved end point. The intensity and duration of exercises in a systematic training program can be precisely controlled and adjusted incrementally. Therefore, a systematic program is repeatable, yields predictable results, ensures adequate rest, and provides an opportunity to evaluate the training protocols used.

A training program is *disciplined* when it is executed dutifully in accordance with the plan, and diligently tracked over time. This attribute ensures that the intended benefits of the training have an opportunity to materialize, and enables evaluation and improvement of the training program in the future.

Finally, a training program is *science based* if it draws from the existing body of knowledge to design the exercises, and is inquisitive in nature. While the ultimate goal is to improve climbing performance, an ever-present, intermediate goal is to *improve the training* by following the Scientific Method: hypothesize, experiment, observe results, evaluate the hypothesis, and repeat. This is necessary because even the best training plan in a *well-studied* sport will produce different results for different athletes, so the athlete or coach must constantly evaluate and adjust the training to achieve the best results. Therefore, in climbing, a relatively new, *under-studied* sport, it is even more critical to apply the Scientific Method to adapt the training plan to best fit the needs of the athlete.



An athlete in *training* sets goals at the beginning of a season, then designs a specific exercise program to reach those goals. The athlete in *training* then follows this pre-planned schedule of workouts throughout the season, diligently documenting the work accomplished, specific results, and overall progress. The athlete in *training* understands the genesis of the exercises he or she is performing, how they should affect his or her body, and what changes might be made to get different results. Finally, the athlete in *training* analyzes the effectiveness of the program and, if necessary, refines it for future benefit.

DO ELITE CLIMBERS TRAIN?

Many climbers apply a spontaneous, rather than systematic, approach to their climbing. They typically adopt this approach early on and are never presented with a good reason to train systematically. This casual method is often reinforced by the impression that the most accomplished climbers follow the same easy-going approach. Be they the local honemasters, or international celebrity rock stars, it is reasonable to suppose that if the most-accomplished climbers don't need to train, then neither does anyone else. However, this argument is problematic for two reasons. First, many top climbers benefit from favorable circumstances that may not be an option for the typical climber. For example, perhaps they started climbing (or gymnastics) at a very young age, and so were able to develop tremendous finger strength during their formative years while their body was most responsive to the unusual stimuli of rock climbing. Others may simply have superior genetic gifts, or a freewheeling lifestyle that allows for more time on the rock to refine their skills. Second, and most importantly, chances are most of these accomplished climbers actually do train, though it's not always obvious. A lot may be going on "behind the scenes" between the few times a month they are *performing* at the gym or crag.

This is the case for many professional climbers. They train hard in private, but this isn't newsworthy, or terribly interesting to the public at-large, so it isn't publicized as much as their seemingly spontaneous forays to the crag where they "just climb" (...everything at the crag). There is a peculiar social norm among climbers that one should create the illusion that he or she doesn't care "too much" or try too hard. This compels many elite climbers to downplay their training, or describe it as something other than "training." Nevertheless, they are training. Dig into their blogs, or speak to them in person, and it is clear.

They may do much of their training on rock, and thus avoid tools such as finger and campus boards, but they are training in the sense that they focus on specific exercises to develop specific capabilities and overcome specific weaknesses in order to accomplish specific goals. It may be an annual three-week pilgrimage to Hueco Tanks to boost finger strength for a particularly bouldery 5.14d, or a pit stop at the Red River Gorge for some endurance work before a trip to Spain, but it *is* training. Unfortunately, most climbers lack the freedom to train in this manner. These climbers can benefit from time-efficient systematic training to achieve similar results. That is what this book offers.

THE CLIMBING ATHLETE

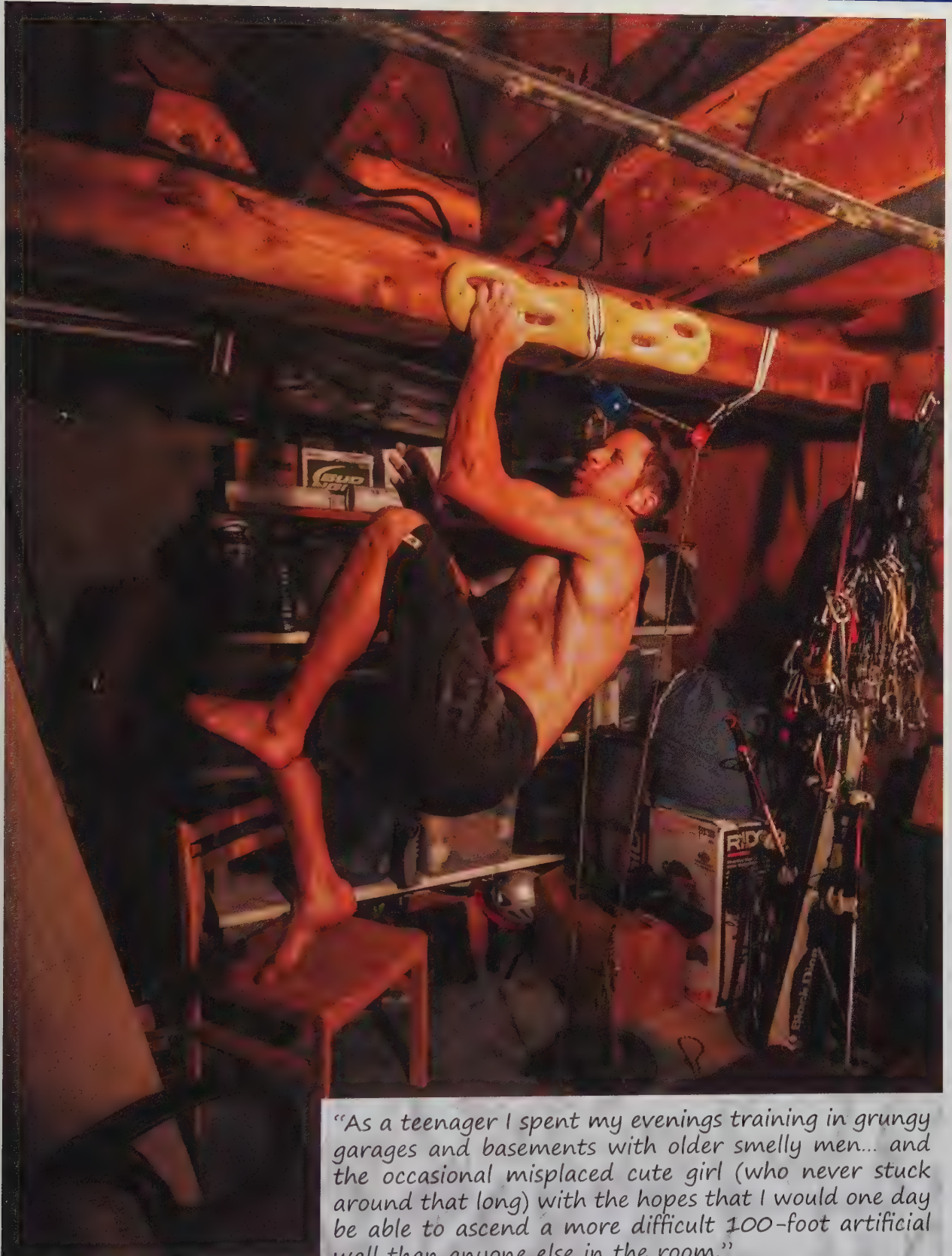
Climbers that are contemplating whether or not to train should consider why they climb and what they want out of it. For *recreational climbers* who climb largely for less-quantifiable reasons such as building camaraderie and visiting interesting places, training won't likely add to their fulfillment. The recreational approach is perfectly acceptable; virtually all climbers (this book's authors included) started out as recreational climbers, and even the most hardcore grade-obsessed cave troll adopts the recreational philosophy from time to time. However, other climbers consider themselves *athletes*, with climbing as their sport. Whether they clip bolts or place pro, they are driven by a desire to push themselves to their ultimate potential, and they have an unquenchable thirst for continuous improvement, like athletes in any sport. Athletes practice and train for their sport, be they cyclists, distance runners, or body builders. It doesn't matter if they are amateurs or pros—athletes train.

"I'm always going to try to improve and I'll always be 'that guy' who requires training, because if I don't I start to suck really bad, really fast."

— Sonnie Trotter⁷

Training is occasionally derided by some climbers as "trying too hard" (which is silly nonsense) or "taking the fun out of climbing." While fun is a matter of personal taste, most climbers with experience on both sides of the fence will attest that training can be fun in its own way, and nothing is more fun than crushing once-hard routes with relative ease.

Something worth doing is worth doing well, and this applies even more so to climbing because of the increased access that results from improvement. If



"As a teenager I spent my evenings training in grungy garages and basements with older smelly men... and the occasional misplaced cute girl (who never stuck around that long) with the hopes that I would one day be able to ascend a more difficult 100-foot artificial wall than anyone else in the room."

— Tommy Caldwell⁴

INTRODUCTION

one derived no other benefit, at a minimum, improved performance increases the pool of climbs available to choose from, and the harder lines are almost always the *best* climbs. A stroll through a typical sport crag should convince any climber that the most spectacular, highest-quality, psyche-inducing lines generally lie on the upper end of the difficulty spectrum. With hard work and training, those climbs are accessible to anyone, along with the incredible satisfaction that comes with sending a dream route. Thus by improving, not only will you be *climbing well*, you will be doing it on *better* climbs.

Improving is the obvious reason to train, and most climbers want to improve. Many climbers believe that they can improve by simply climbing as much as possible. That is true to an extent. A certain amount of improvement can be had simply through random climbing. Physically, there will be improvement so long as the volume of climbing is steadily

increased (something that is hard to accomplish randomly). Technique and other less tangible factors will improve initially as well. With the availability of indoor climbing facilities and access to great sport climbing, it is possible to improve to a point without systematic training (though this still requires climbing two to three times per week). A reasonably gifted climber might reach a high level of expertise by “just climbing,” but will eventually *plateau* (stop improving). For some folks, it is acceptable to reach an arbitrary grade and hover there, off into the sunset. For most, this is perpetually frustrating. So why not “just climb”? There are several reasons:

CONTINUOUS IMPROVEMENT

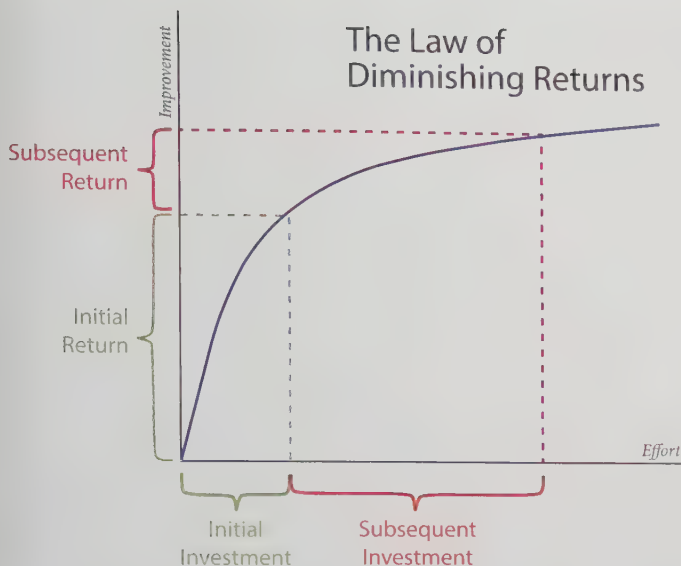
The first reason to train is that the best way to *ensure* steady, continuous improvement is with training. Athletes in all sports experience improvement to a point, and then eventually, long, frustrating performance plateaus in which their level of performance is constant and they don’t experience any improvement. Anyone who has been climbing long enough is likely familiar with this. With a random “just climb”-type of routine, there is no way to force improvement because there is no record of what work was accomplished to reach that point, and therefore, no baseline from which to incrementally increase the load on the body in order to force adaptation. As monotonous and mundane as training can be, at least the results will not be monotonous. Improvement will come, and the training record will reflect that improvement, even if the circumstances of a given climbing season don’t allow new “sends.”

The knowledge that one is getting stronger can be very motivating, and can carry a climber through a number of otherwise disappointing seasons. With the “just climb” approach, the only evidence of improvement is with a “send.” Imagine trying to lose weight without stepping on a scale; it would be nearly impossible to stay motivated because there would be no evidence that the sacrifices involved were paying off. Sometimes life circumstances, be they work, school, a new friend, a new baby, or even just bad weather at the crag of choice, can prevent sending, whether the climber was due for better performance or not. With a training record, the simple knowledge that gains were made during training can sustain motivation through to the next season, when there will be an opportunity to “send” again.



Often a crag’s best routes are also among the hardest, inspiring us to raise our game in order to climb them. Mike Tritt stoning *Goliath*, 5.13a, at Enchanted Tower, NM. 📷 Jason Hundhausen

The Law of Diminishing Returns



INJURY PREVENTION

The second and most compelling reason to train is injury prevention. Stop and catch your breath—that wasn't a typo. Yes, training *prevents* injury. Here's how: Imagine two athletes lifting weights. Athlete A exercises according to no specific schedule whatsoever. He lifts whenever he has time, wherein he does one squat rep of a random weight then a couple reps of bicep curls (again at random weight, and so on), then a pull-up followed by two more squat reps, then does one max rep on triceps extensions, then does a bicep curl then one max rep on leg extension then one military press, then maxes out on bench press, then three sit-ups, etc. Suppose he does this for three straight days because it's a holiday weekend and he has the free time.

On the other hand, Athlete B goes to the weight room according to a schedule she has pre-planned that allows for hard days, easy days, and rest days. When she's in the gym she warms up each muscle group before working it, then she lifts a specified weight that is based on what she has lifted during her previous workouts plus an incremental increase (say five percent), closely monitoring her body's response to the load. Which athlete is more likely to get injured?

While this is an extreme example used to make a point, most climbers resemble Athlete A. Their typical workout is spontaneous indoor bouldering or route climbing, which is essentially a series of random exercises of various intensity and duration, sporadically involving various muscle groups. It offers few options to isolate and precisely stress the muscle (and supporting structure) to a specified amount, and there is no

way to measure the workload. It is virtually impossible to know if the intensity is increasing, or by how much. Unfortunately these sessions occasionally end in a wild dyno for some tweaky hold that the climber is not prepared to hold onto, then the telltale "pop" of torn connective tissue.

This spontaneous approach to improving contradicts the most fundamental rules of sport science and physical development: that the optimal way to stimulate muscular adaptation is to progressively stress it to near-failure. Furthermore, simply because a workout induces fatigue doesn't mean it will stimulate the desired physical adaptations. With random bouldering, there is no way to know when that *near-failure* point is reached, or if the desired *type* of near-failure has been achieved. However, it is always clear once that point has been exceeded (injury). The support structures of the fingers (tendons and ligaments) require up to *six years* to adapt to stress. Therefore a climber of six years is just now developing the additional ligament strength needed to handle the stress of hard climbing. Forearm muscles respond to stress in about two weeks. Therefore, within weeks of becoming a climber, the forearm muscles acquire the ability to destroy the support structure. On the other hand, a structured training regimen allows the climber to carefully and thoughtfully stress the body enough to cause the desired adaptation without stepping over the line into injury. Aside from giving up climbing for fishing, training is the best way to avoid injury!

REHABILITATION

Injuries might still happen from time to time, but once a climber adopts the mindset of an athlete in training, injuries will rarely hold him back for long. An athlete in training becomes an expert in listening to his body, identifying weakness, stressing weak tissue to stimulate growth, and managing rest periods to maximize recovery. Therefore, the training process of carefully applying stress to the body to maximize adaptation creates an understanding for exactly how much stress the weaker structures can handle. Training will also introduce the athlete to a variety of tools to use in the event rehabilitation becomes necessary, while providing a thorough historical strength baseline from which to develop rehabilitation plans and to track recovery. Not only does training reduce the risk of injury, it also provides a methodology and the background data necessary to facilitate recovery from injuries, should they occur. *Training* and *rehabilitation* are just two sides of the same coin.

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TRAINING SAVES TIME

Finally, a third reason to train is that it simply takes less time. The systematic nature of training makes it the most efficient way to completely and thoroughly exercise. Nothing is more *inefficient*, in terms of time, than climbing an assortment of routes at the crag in hopes of fatiguing all of the relevant muscle groups in order to stimulate adaptation. If it were even possible to reach total exhaustion with this “just climb” approach, it’s fair to say that it would require several more hours per session. Through systematic training, weak areas that need stimulation can be precisely identified, and the appropriate stress can be applied exactly where needed with very little wasted time and effort. Every grip position can be precisely stressed to failure in less than 90 minutes with the proper equipment and a well-designed program. Plus it can be done without a partner, at the gym or at home, any time, any day, regardless of weather, work schedules, etc. Good luck getting that at the crag! For those who don’t have 40 hours per week to climb, training is by far the most time-efficient way to improve; for working adults with families, it’s the *only* way to improve.

TIME VALUE OF CLIMBING ABILITY

Economists use the concept of Time Value of Money to explain the benefits of financial investment, and a similar case can be made for the benefits of training. The concept is that \$100 invested today will be worth \$105 a year from now, assuming an annual net interest rate of 5 percent, at which time the investor will have more buying power than he has today. There is a similar Time Value of Climbing Ability, in that *time* invested today in improving climbing ability will yield more sending-power in the future.

For example, consider a climber who consistently redpoints 5.12c in six tries, 5.12b in three tries, and onsights 5.12a about 50 percent of the time. Supposing she gets three quality redpoint attempts per climbing day, she should be able to redpoint a given 5.12c in two days of effort. If she saves a few climbing days each month to invest in following a structured training program, in a year she might be redpointing 5.12d in six tries, 5.12c in three tries, and onsighting 5.12b about 50 percent of the time. The two days she required to climb *one* 5.12c a year ago will now get her *two* 5.12c routes — more *sending-power* and a great return on her initial investment.

Many climbers skip training because they are so eager to climb every waking moment that they can’t see past the routes right in front of them. However, a

small up-front investment of time and effort toward training will yield improvement that will easily make up for the temporary delay in the immediate pursuit of sends. Granted, this analysis is simplistic; quality of life is not measured simply in terms of number of sends per day, and a harder send is not necessarily “better.” However, taking the long view and accepting that every climber’s time is limited, why not create the option that makes the most of that time? A training program that provides steady, reliable improvement is the best way to maximize climbing potential over a lifetime.

TRAINING IS FOR YOU!

Many climbers lead hectic lives, and so believe they lack the time to train due to other commitments, be they a stressful job, family, school, or other pursuit. The approach proposed here is ideal for these climbers because it offers the most *time-efficient* training program available. With the myriad climbing exercises out there, and the volume of activities going on at a local gym, it would be easy to assume training requires a lot of time, but it doesn’t have to. This book describes a number of “luxury” exercises and activities so that those with extra time can strive to master their craft, but these activities are not intended for everyone. The most essential activities will be clearly identified so they can be prioritized to achieve the greatest possible benefit commensurate with the amount of time committed. The most critical exercises are *training dense*, packing a great deal of training stimulus into short time periods, yielding maximum improvement with the least cost to everyday life. What’s more, most are possible to do at home, any time of day, so the lengthy commute to an elaborate climbing gym can be avoided. Those who feel that they are too busy to train may in fact be too busy *not* to train! For the very busy, what little time is available for climbing is that much more precious, and so it is that much more critical to make the most of that time.

MANY PATHS, BUT THIS ONE WORKS FOR MANY PEOPLE

When studying the training and climbing habits of elite climbers, it becomes apparent that there are many paths to success in rock climbing. If the world’s 100 best climbers were asked how to train, there would be 100 unique answers. Some train like fiends, eight hours per day, seven days a week, to reach the same fitness level as those who simply “go climbing” a few days per week. The reasons for these disparities may

Family, Career, and Climbing — Is it Possible to Have It All?

In a perfect world we would all have 20-plus hours a week we could spend training, rehearsing technique in a state-of-the-art gym, or climbing outdoors on wonderful sport routes. Unfortunately, most of us can only dream of that. At the most hectic times of my life (when my kids were first born, studying for a PhD, etc.), I prioritized my training time for maximum effect. That meant leaving out many beneficial activities, but it also allowed a crucial balance. I kept up the most-critical training activities such as finger-strength training, and in so doing was able to maintain my level of performance through chaotic life events that might lead other folks to give up climbing altogether. As these events passed and I had more time, I reinstated many “luxury” training activities and, when combined with the “critical training tasks” I had maintained through the chaotic times, I saw massive improvements. While high-level climbing often involves choosing to skip out on more traditional features of life, such as a career, house, and a family, it doesn’t *require* that choice. There are plenty of examples of folks who “have it all,” but it *does* require hard work and careful prioritizing for efficient training.

— Mike Anderson

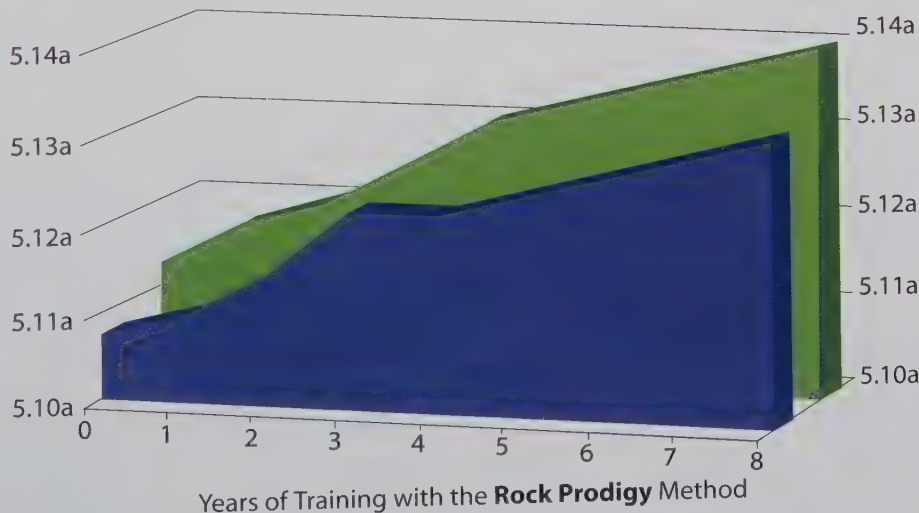


Time-efficient training makes it possible to improve despite a busy family and professional life. Mike Anderson sent his first 5.14, *God's Own Stone*, with two kids in tow and a PhD in the bag. 📷 Anthony Carco

never be known. The problem then becomes how to select a training program with so many different options that all seem—at least to some extent—to work.

Any given program is probably better than some and worse than others, with the athlete as the primary variable in determining relative success. Any program should ultimately be tailored to the individual athlete, but learning how to tailor a training program takes wisdom and experience. Thus it makes sense to start with a basic program that has a broad, proven track record of success. If it’s necessary to be a genetic freak or superstar athlete simply to complete a single routine, it’s probably not a good choice for most folks. Similarly, if a program depends upon superhuman (or even teenage-human) recovery

abilities to survive the workload, it’s probably a worse choice. Many programs from climbing’s elite prescribe loads and frequencies that would cripple the rest of us. Other programs rely on the inherent genetic talent or circumstantial advantages of the athlete, so while they work for some they are ineffective for most.



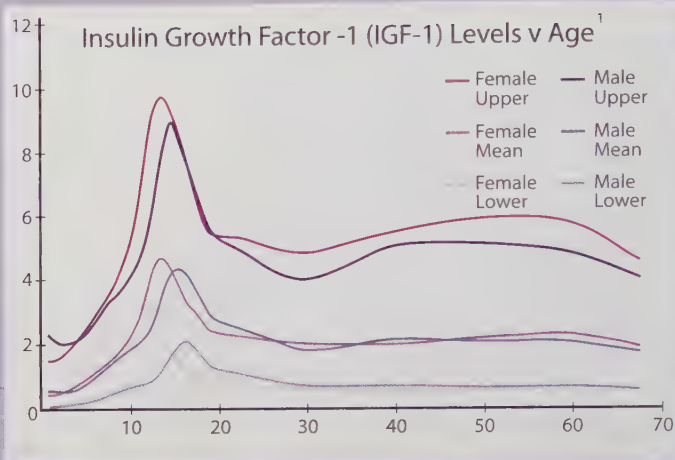
The “Wunderkind” Method

“I’ve never really had a problem with injuries, as my fingers are naturally strong. Then again it’s probably because I did so much finger training at that critical age between 14 and 18. Finger strength is everything, and if you lay down a foundation at that age then it gives you a real advantage, both in terms of performance and avoiding injury.” — Malcolm Smith⁶

A common pitfall in planning a training program is to co-opt the program of a climbing idol. Unfortunately, this strategy rarely works. The idol’s methods are not necessarily incorrect, but they are specific to his or her abilities, talents, and circumstances, and do not transfer well to the general community. Consider the climbing “wunderkind.” He or she starts climbing as a child, frequenting the local gym, and eventually crushes all the local testpieces. So why not mimic the training approach of the young star? *Usually* the climbing volume

of these whippersnappers would cripple the typical adult, even young adults (those in their early twenties). The “growth spurts” that occur during and immediately after puberty permit extreme durations and intensity of exercise while facilitating rapid recovery. Adult bodies cannot handle this workload or recover as quickly. In addition to their seeming invincibility, the blessed young require relatively little stress to stimulate adaptation, so don’t assume that the activities shown to stimulate growth in a teenager will produce the same effect in an adult.

It’s reasonable to argue that because Adam Ondra is the greatest climber in the world, climbers should simply do as Adam did to reach the pinnacle of the sport. However, Ondra started climbing at age six, with parents who were avid climbers and could guide his development. Without a time machine, adult climbers simply cannot do what Adam did (that is, start climbing at age six). It’s too late to leverage the physiological benefits of puberty to develop superhuman finger strength. While aspects of Ondra’s training may be beneficial to the larger climbing community, the fundamental key to his success is not an option for the vast majority of climbers. Instead, seek a training program that is safe for adults, and works for adults. Compensate for a late start in climbing by utilizing every other advantage available, such as the heightened wisdom, patience, and perseverance that come with maturity.



The program described here may not be better than any of the others, but it has produced excellent results (often dramatic results!) for a wide variety of climbers with different athletic backgrounds, weaknesses, and abilities. It works for those who climb 5.10 and those who climb 5.14. It has continued to produce consistent improvement even for athletes who have followed the program for 10-plus years. It has propelled numerous climbers, with different genetic gifts, to the 5.14 level. It may not be the best possible program, and it’s certainly not perfect, but it is one path to success that is proven to work.

PATIENCE AND PERSEVERANCE

The case for training has been made in earnest, assuming, perhaps incorrectly, that many climbers need to be convinced. On the other hand, it is possible to be *too eager* to try a new training routine. Some folks are so eager to improve that they religiously scour climber blogs for ideas, and then experiment with whatever looks promising. The danger with this approach is that whatever training was already in progress is aborted prematurely to make way for the latest “flavor of the month.” Tragically, even if an effective approach comes along, these eager beavers are unlikely to no-

The Send-less Seasons

You trained your backside off all winter, making countless sacrifices, certain that they would pay off with breakthrough sends in the spring, but they never materialized. Entire seasons can pass “send-less” simply from picking the wrong project, or by squandering fitness on a route that is too easy, too hard, or otherwise fails to coincide with the training. That doesn’t mean that the training activities are flawed or that the season was wasted. Improvement at such a complex sport takes time, and often, the timescales of climbing improvement don’t coincide with our expectations or fit neatly into our life plan. That letter-grade jump you feel you deserve might need two or more seasons to manifest itself with a send. Nevertheless, a send-less season can be *very* productive.

I spent the entire summer season of 2011 working a single route. After weeks of training, significant sacrifices, and eight days spent on the route, I suffered a season-ending skin injury and left the project empty-handed. However, my improvements that season were clearly documented. I achieved personal bests on my hangboard, campus board, and interval circuit. Additionally, I made significant progress on a route that seemed well beyond my ability at the outset. I knew that I had reached a new level, even though the season produced no significant sends.

With new confidence and tremendous motivation, I entered my next training cycle absolutely determined to finish what I had started. Although that send-less summer season is now virtually invisible on my tick-list, it led directly to my eventual success on *Grand Ol’ Opry* the following season. In the end, that “send-less season” turned out to be one of the most productive seasons of my career.

— Mark Anderson



Mark Anderson redpointing *Grand Ol’ Opry*, 5.14c in the fall of 2011, at the Monastery, Colorado. 📷 Matthew Oscadal

tice because they don’t give it time to pay off. In order to evaluate the effectiveness of an exercise, or more importantly, a training program *as a whole*, it is necessary to follow it consistently through to its conclusion. It may take multiple *seasons* to draw a fair conclusion. Yet some climbers make this mistake routinely by

adopting the next training method too quickly, before the current method has had time to produce results.

The program described on these pages works best when it is followed thoroughly to completion. This cannot be overstated: *The program relies on synergies that are designed to be effective only when*

the program is followed in its entirety. It is possible to cherry-pick specific workouts to achieve specific results, but if you do so, less-than-optimal results should be expected. Depending on the circumstances of a given climbing season, the program may need to be followed for multiple seasons before the improvement materializes into a personal best, but it *will* come eventually. A lot of serendipity is required to have a magical season of tangible progress, and any number of circumstances can derail it (bad weather, lack of climbing partners, crag closures, illness, etc.). When this happens, as it does from time to time, it's very hard to identify a singular root cause (such as a particular training activity), because success in climbing depends on so many variables. Don't rush to judge a training program based on one unproductive season. Every training program will yield unproductive seasons. The question is whether or not the program will eventually yield progress. For those who diligently follow the program, it has never failed to produce results *eventually*, though success requires perseverance through the occasional "bad" season. It is unrealistic to expect a letter grade of improvement every season (if that were possible, 5.19d onsights would be commonplace). Climbing is extremely difficult to master, and the longer one climbs and the longer one trains, the more difficult it is to eke out tangible improvement. Even for those following the "perfect" program, improvement still requires tremendous effort. Dramatic gains can be achieved initially with less rigor and effort, but over time, greater effort and intensity will be required to maintain improvement. Robotically following the prescribed workouts is not enough to improve indefinitely in such a complex sport. Strengths and weaknesses must be ceaselessly scrutinized, and time and effort must be applied to practicing away your weaknesses.

Furthermore, training does not remove the requirement to *try like hell* at the crag. Individual moves and routes will still be hard. Training bestows the physical and mental abilities to apply tremendous effort to such moves, and sustain that effort on a route, but it doesn't make hard moves feel easy. Finally, transferring training from plastic to rock is not trivial. Making the transition requires conscious thought, some patience, and lots of focused effort. This book will address ways to accomplish this in **Chapter Three: Skill Development** and in the performance chapters in Part III of the book.

"If you are learning to climb from the guy/gal in the gym with a big university degree but little common sense, run away. You can't put a price on mileage. Try to find a teacher or coach who can offer you both: a big brain and an even bigger box of life/climbing-related experience. You won't be disappointed."

— Sonnie Trotter⁸

WHY THIS PROGRAM?

Though climbing is one of the oldest and most natural sports, extreme rock climbing is a very recent development, so it has not been studied very well. As a result, there is a limited body of knowledge available to climbers, and most of what is available has merely been adapted from other, similar sports. Unfortunately, there isn't another well-studied sport that is very similar to climbing. Many theories are taken from powerlifting and middle-distance running, and recently, dance. It's not that these sports are the most closely related to climbing in terms of physical demands on the body, but rather, they are considered the most similar *among sports that have been thoroughly studied*. Researchers occasionally perform climbing-specific studies, but for the most part these studies are limited due to lack of resources and facilities, resulting in little useful new information. Thus for the time being, it is up to amateur climbing scientists to do their own informal experimentation.

Given the lack of a respected establishment to tell climbers how to train, it is wise to question sources. Whose advice should be followed? In more established sports, coaches are vetted over a long career, and the very best percolate to elite, international competitions. Expertise is not so obvious with climbing, at least not in America. Instead, climbers are self-coached, so we usually seek advice from successful climbers. This can be a big mistake, however. Douglas Hunter, a successful coach himself and the author of *The Self-Coached Climber*, has admonished people, "not to assume that an elite climber knows why he or she is elite." For example, a World Cup-level climber could probably spend the next year carving pumpkins as his "training," and he would still climb better than most, but it doesn't mean pumpkin carving is the best training. How hard someone climbs at a point in time is not enough information to evaluate the quality of his or her training. Rather, expertise should be judged

based on *improvement over time*. A 5.13a climber who has been stuck at 5.13a for five or 10 years may have stumbled upon a dead end that other climbers would do well to avoid. On the other hand, a climber currently at the more modest grade of 5.12c, but who was climbing 5.11a two years ago and has been making steady progress, is probably doing something worth imitating.

Effective coaching requires three main characteristics: theoretical knowledge, practical experience, and innovative thinking. This theoretical knowledge entails an understanding of how the body works, including the biomechanics of climbing movement, the biochemical processes of muscles, and general principles for forcing physical adaptation. This knowledge and experience enable the coach to determine which mental, technical, and physical attributes need to be improved to achieve a particular goal, and which training protocols are likely to stimulate the necessary adaptation. However, this theoretical knowledge is only useful to climbers when it culminates in a practical training program that produces results on the rock.

Practical experience is perhaps the most difficult coaching trait to come by. Theoretical knowledge is plentiful, but it is not enough by itself. Theoretical knowledge is almost always based on non-climbing-specific research and relies heavily on assumptions and leaps that should not be accepted as fact until

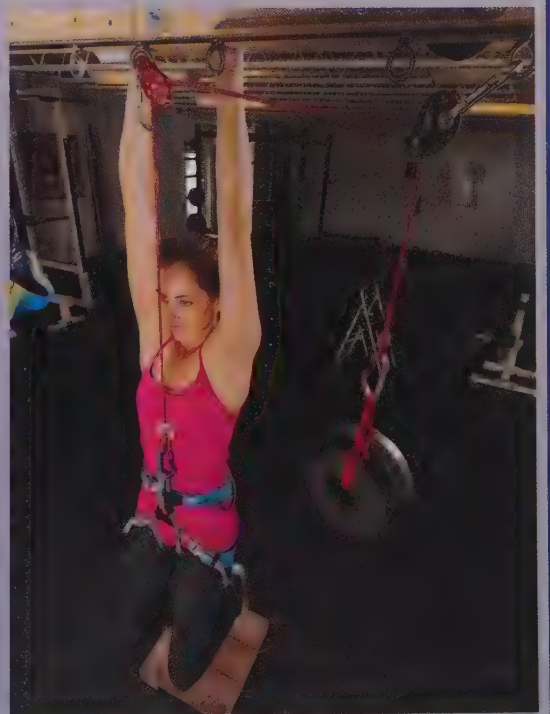
demonstrated on actual athletes in a realistic environment (like, for example, that human muscle behaves the same as dead-rat muscle). Practical experimentation is required to determine how the body will respond to climbing-specific exercises, and whether such a response will actually improve climbing performance. The muscles used in climbing, and the constraints on them, are so unique that one can't simply transfer techniques from any sport to climbing. For example, the finger flexors are almost always used isometrically in climbing, but there is little theoretical knowledge on the best way to train isometric strength because it is rarely required by other, well-studied sports. The phenomena that cause a forearm pump are even more one of a kind. Therefore, the only way to obtain climbing-specific knowledge is through practical experimentation.

Finally, innovative thinking is critical because climbing is such a fresh sport. The exercises climbers use today such as hangboarding, campusing, and intervals were all invented within the last two decades by thoughtful climbers looking for an edge. The perfect training exercise may be yet to be discovered. The process of training innovation requires analyzing the skill or physical adaptation that must be developed, imagining an exercise that will pinpoint the weakness, and then inventing the apparatus that enables that exercise.

Training Innovations

When we first started hangboarding, the existing guidance was to put your feet on a chair to lessen the intensity or load on the fingers. This technique is highly variable, unquantifiable, and thus completely inadequate. The next suggestion was to stand in long rubber bands to subtract weight. This was a slight improvement, but still too variable and only crudely quantitative (Google "Hooke's law" for more). Instead, we developed a pulley system to remove weight precisely and accurately. Though it may seem obvious now, it wasn't at all obvious at the time, yet this simple innovation enables the development of finger strength on the most heinous (and most critical) holds, something that might never have been possible if we left our feet on the chair.

— Mike Anderson



With an understanding of what makes effective coaching, each climber should work to become her own great coach, which means studying the body, experimenting, and innovating. The authors started by adopting the theories of pioneers like Gullich, Neumann, and Yaniro, but since then the program has evolved through personal experimentation to test out new ideas, and thus been refined. It takes years to test new ideas on one person, but as more climbers become interested in training, collective experimentation will yield exciting new advances. Furthermore, the human body is a complex machine. Everyone is different, and everyone will respond to a specific training program differently. Therefore, it is incumbent upon each climber to be flexible. Start with the

basic program, pay close attention to your body's reaction, and make corrections as needed. Try different rest periods, or different numbers of reps, sets, etc. That is how new knowledge is discovered.

THERE IS NO TIME LIKE THE PRESENT

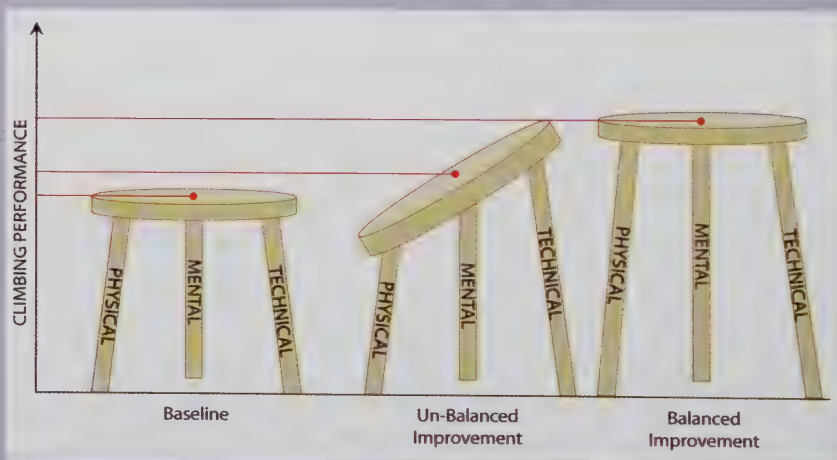
Given the veritable heap of discarded training programs lying about, it would be easy to adopt a "wait and see" approach to training, continuing to "just climb" while waiting for the "perfect plan" to materialize. If this sounds appealing, it would be wise to reconsider. The legendary World War II General George Patton once said: "*A good plan now is better than the perfect plan next week.*" While climbing is not war, the principle applies. This book details what is, at the very

Components of Climbing Performance — A Three-Legged Stool

Overall climbing performance is a combination of physical, mental, and technical components. The relative importance of these components varies according to the demands of specific routes, and it also varies throughout a climber's career. For example, the performance of a novice gym climber is typically dominated by her technical competence, while physical and mental factors are largely irrelevant. An experienced sport climber learning to trad-climb may be limited primarily by mental factors. For an experienced climber on a steep power route, the physical component may be more limiting. Nevertheless, these components all play an important role, and a deficiency in any one component can hinder a climber's overall performance. It is helpful to think of climbing performance as the top of a stool, supported by three "pillars": physical, mental, and technical. Total climbing performance benefits the most from growing all of the pillars in unison. It is possible to focus on one pillar while neglecting the other two, and still see an improvement in performance for a while, but that approach prevents us from reaching our true potential.

Fortunately for climbers, these components are not mutually exclusive. It is possible to construct a training program with complementary activities that benefit each of the three pillars, without ignoring the

others. The Rock Prodigy method is just such a training program. The program includes ample time and numerous activities that can be used to develop and improve your technical and mental climbing abilities. In **Chapter Three: Skill Development**, we thoroughly describe effective techniques for skill development, and how they should be integrated into the training program. In **Part III: Performing**, we present numerous strategies for mastering the mental aspects of performance. Utilize these methods to bring your pillars of performance into balance.



The three pillars of climbing performance. Growing all three pillars in unison is the best way to improve total climbing performance. It's difficult to reach for the stars when standing on an unbalanced platform.

least, a “good plan.” Beginning with this immediately and sticking with it will put any climber far ahead of where she might be if she waited for that perfect plan. Time is a climber’s most valuable commodity, and waiting for something better squanders it. Don’t wait: Start now and be all you can be, right away!

THE ROCK PRODIGY TRAINING METHOD

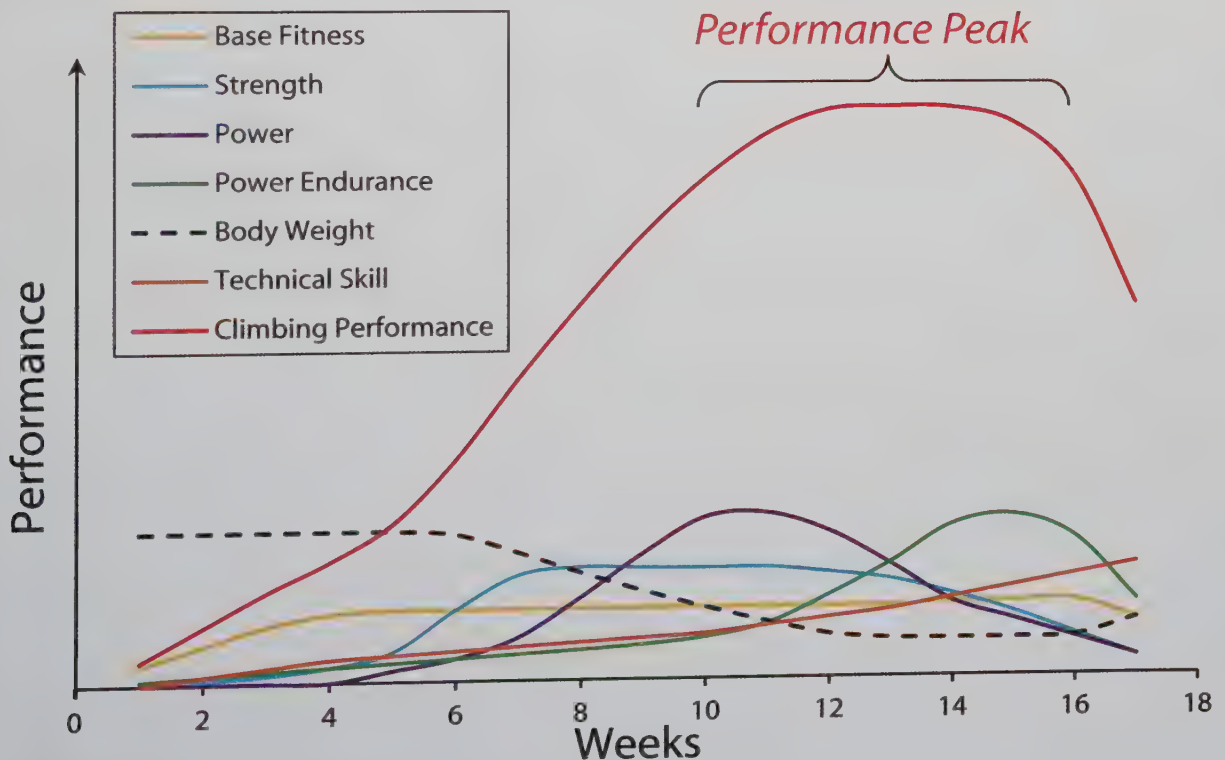
The program described in the following chapters was originally developed by Mike Anderson in the late 1990s, based on Neumann and Goddard’s book *Performance Rock Climbing*² and Petro and Yaniro’s training video *Fingers of Steel*³. Since that time, the program has evolved substantially to incorporate advances in sport science, personal experimentation, years of experience, and to broaden the applicability to climbers in different stages of development.

The Rock Prodigy training method is a form of *periodization*—a strategy for physical training in which exercises are carefully varied to avoid plateaus and create synergies that result in a *performance peak* at a predictable time (see figure below). Climbing requires diverse fitness, for example, *power* to execute

extremely difficult single moves, along with *endurance* to sustain such moves. Through careful timing, the periodization of the Rock Prodigy method enables climbers to simultaneously develop a high level of each aspect of fitness. The combined effect is the performance peak: a temporary, but significant boost in climbing fitness that will enable the climber to attain never-before-seen levels of performance. The basic framework of this program is a 17-week season consisting of the following six training phases:

- Base Fitness (lasting two to four weeks)
- Strength (three to four weeks)
- Power (two to three weeks)
- Power Endurance (two to four weeks)
- Performance (three to four weeks)
- Rest (one to two weeks)

The length of each phase is designed to promote adaptation continuously up to the verge of a plateau, at which time the phase ends and a new phase begins. Phase and season length can be varied to meet the specific needs of each climber. The performance peak typically lasts six to eight weeks, during which time the athlete will climb better than ever and can plan on climbing as much as possible to reap the benefits of the season’s training.



The Effects of Periodization. By timing the training of each fitness attribute, a substantial Performance Peak can be created.

THE ROUTE AHEAD...

If you're ready to raise your climbing to new heights, this book will show you how! The book's chapters are divided into three main parts:

- Part I: Taking Action
- Part II: Physical Training
- Part III: Performing

Part I: Taking Action presents techniques for setting climbing goals and strategies for achieving them through physical training and skill development. **Part II: Physical Training** describes how to hone your body into the ultimate climbing machine! The theoretical foundation for the Rock Prodigy program is presented, followed by detailed training instructions for developing Base Fitness, Strength, Power, and Power-Endurance. Part II also explains how and when to rest in order to prevent injuries, and how to rehabilitate injuries should they occur. Next, detailed training calendars are included that eliminate the guesswork by showing you exactly what to do and when. For more experienced climbers, instructions are provided for modifying the calendar to achieve different results. The final chapter discusses effective techniques for weight management that, when coupled with the prescribed training, will raise your climbing performance exponentially.

Once you've transformed your body into the ultimate climbing specimen, **Part III: Performing** takes you out to the crags. These chapters describe the cutting-edge tools, strategies, and tactics used by elite climbers to slay their dream routes. Discussions include how to select a climbing project, how to efficiently learn and rehearse the moves, how to prepare mentally and physically for the send, and how to stay focused and relaxed to realize your climbing potential. Detailed, step-by-step instructions that are both effective and time-tested are given for approaching redpoint projects and onsight attempts. Finally, the unique aspects of traditional and big-wall free climbing are discussed, along with many never-before published techniques for succeeding on these demanding adventures.

If your palms are sweating, then this is the program for you. Now is the time to rise to the occasion and take action to improve your climbing — the belay is on. So lace up your climbing shoes and get ready to send it!!!

Physical Training Theory — An Acquired Taste

Many of the chapters in **Part II: Physical Training** include detailed descriptions of human physiology and sports science that serve to explain the genesis of the physical training exercises described. Knowing *why* activities are prescribed will increase your confidence in the training program, and thus also increase your motivation to follow through with the training details. Also, knowing *why* gives you the understanding necessary to modify the program to fit your unique body, weaknesses, and climbing goals. In short, it empowers you to customize your training and “coach” yourself. If you're eager to start training and don't wish to wade through this theory right away, feel free to skip ahead to the juicy training details. Understanding the theory isn't required in order to succeed with this program, and it will always be waiting if you become curious at some point in the future.

QUICK-START GUIDE TO THE ROCK PRODIGY TRAINING PROGRAM

The Quick-Start Guide (See Figure on next page) is for climbers who are so eager to train that they can't bear to read this entire book first. It will get them started on the path without delay by highlighting the essential topics needed to execute the basic Rock Prodigy training program. To use the guide, start with Step One and select one of the *Seasonal Training Plans* referred to in the central circle (these plans describe *when* and *what* to train). The brief passages indicated in the steps that follow describe the training activities (how to train). Perform these exercises according to the chosen *Plan*. Refer back to the *Seasonal Training Plan* before beginning each step in the program. For full value, read the rest of the manuscript once your training is underway.

“The hardest thing is to start training; training itself is easier.”

— Iker Pou⁵

Rock Prodigy Quick Start Guide

START HERE

Step 1: Develop a Plan

- The Rock Prodigy Method (Ch4, p81-82)
- Seasonal Planning (Ch2, p37-38)

Step 7: Rest

- Rest Days (Ch9, p169-170)
- The Final Phase, When to Call it Quits, Rest Phase Activities (Ch9, p173-175)

Step 2: Base Fitness & Skill Development

- ARC Training (Ch 5, p96-100)
- Movement Drills (Ch3, p62-67)

Building a Seasonal Training Plan (Ch10, p185-189)

Step 6: Performance

- The Performance Process (Ch12, p229-233)
- Strategies for a Short Term Redpoint/ Long Term Redpoint (Ch13, p237-245)
- Performing During an Onsite (Ch13, p253-254)

Step 3: Strength

- Hangboard Routines (Ch6, p118-122)
- Body Strength Training (Ch6, p123-126)

Step 5: Power Endurance

- Power Endurance Training (Ch8, p153-156)
- Linked Bouldering Circuits, Route Intervals & Performing the Workout (Ch8, p157-162)

Step 4: Power

- Power Training Methods (Ch7, p132-136)
- Beginner Campus Routine (Ch7, p141-142)

PROTECTING CLIMBING ACCESS SINCE 1991



Jonathan Siegel climbs the Third Millennium (14a) at the Monastery, Colorado. Photo by: Keith Ladzinski